Tecnologias de Bases de Dados

PROJ1 – Delivery deadline: 06 April 2026

CODE EFFICIENCY / QUERY OPTIMZATION ASSIGNMENT

ASSIGNMENT GOALS

Analyze the execution plans of different SQL queries executed over a test database. Evaluate the impact of indexes, statistics, and different strategies for structuring queries. The report must include, for each experiment: the SQL query formulation, the obtained result, execution conditions (including execution time if relevant), the execution plan, and the conclusions derived from the analysis.

TEAM

The assignment must be executed by teams of 1 to 2 elements.

SUBJECT

This database stores information about the Portuguese parliamentary elections held on October 10, 1999.

The country is divided into electoral districts, here designated as districts. In the islands, the electoral district coincides with the region. Political parties present candidate lists in the districts where they compete, and mandates (seats in parliament) are assigned at the district level. The emigrant districts are not considered in this dataset, so the total number of mandates is 226.

Voting results for parties are recorded with a finer granularity, at the parish (freguesia) level. Parishes belong to municipalities (concelhos), which in turn belong to districts. All administrative divisions have a code and a name. Districts belong to one of three regions: 'C' – Mainland (Continente), 'A' – Azores, and 'M' – Madeira.

The total number of votes for parties, plus blank votes and null votes, constitutes the number of voters. Adding voters and abstentions results in the number of registered voters. Information about registered voters, abstentions, blank votes, and null votes is only available at the district level.

Political parties may present candidate lists only in some districts.

DATA MODEL

DISTRITOS(<u>codigo</u>, nome, regiao)
CONCELHOS(<u>codigo</u>, nome, distrito → DISTRITO)
FREGUESIAS(<u>codigo</u>, nome, concelho → CONCELHO)
PARTIDOS(<u>sigla</u>, designacao)
VOTACOES(<u>partido</u> → PARTIDO, <u>freguesia</u> → FREGUESIA, votos)
LISTAS(<u>distrito</u> → DISTRITO, <u>partido</u> → PARTIDO, mandatos)
PARTICIPACOES(<u>distrito</u> → DISTRITO, inscritos, votantes, abstencoes, brancos, nulos)

This database has 7 tables populated with real data.

- The **freguesias**, **concelhos**, and **distritos** tables refer to administrative divisions and electoral constituencies.

For each freguesia, it is possible to know its code, its name, and the code of the concelho to which it belongs.
For each concelho, there is information about its code, name, and the code of the distrito (electoral constituency) to which it belongs.
For each distrito, there is information about its code, name, and the region to which it belongs ('C' – Mainland, 'A' – Azores, 'M' – Madeira).

- The **listas** table contains information about the number of seats obtained by each party (acronym) in the districts (code) where it ran.
  The value 0 means that the party ran but did not obtain any seats.
- The **votacoes** table stores the results obtained by the different parties in the freguesias. A null value in votes means that the party ran but there was no vote count (boycott).
- The **participacoes** table contains, for each district code, information regarding the number of registered voters, voters, abstentions, null votes, and blank votes. Note that:
  registered voters = voters + abstentions
  voters = sum of votes in the district + blank votes + null votes
- The **partidos** table stores the acronym and full name of the 12 existing parties.

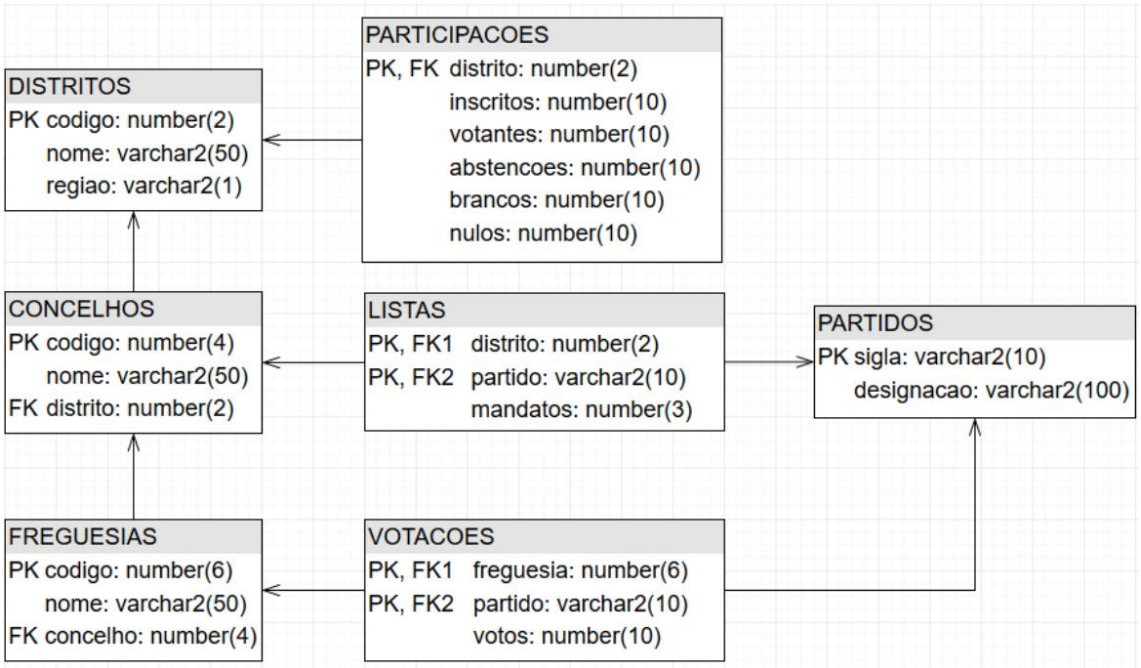

Figure 1 – Relational model schema.

DATA

The tables are located under the GTD7 user, from where they should be copied, in triplicate, into the tables xlistas, xvotacoes, …, ylistas, yvotacoes, …, zlistas, zvotacoes, … . The objective is to create three experimental environments:

X – Without indexes or integrity constraints.
Y – With standard integrity constraints (primary key and foreign key).
Z – With standard integrity constraints and any additional indexes you consider appropriate, justifying the reason for including each of them

METHOD

For the queries below, determine the SQL formulation and the respective answers. Analyze the execution plans and estimate the effort required. The queries must be executed in the three environments (X, Y, Z).

QUERIES TO ANALYZE

1. Selection and Join
    a. What are the codes and names of parishes belonging to municipality 1103? And to municipality 'Azambuja'?
    b. Indicate the party acronyms, designations, and number of mandates obtained in the district of Lisbon.
    c. Indicate the number of votes obtained by the party BE in parishes of the Lisbon district.

2. Aggregation
    a. How many votes did the party PS obtain nationwide?
    b. How many votes did each party obtain in each district?
    c. Which party obtained the highest number of votes in a parish? Indicate the party acronym, parish name, and the corresponding votes.
    d. For each district, indicate its name and the party that obtained the highest number of votes in that district.

3. Negation

    Likewise in question 2, analyze the query: 'Which parties did not compete in the district of Lisbon?'

4. Universal Query

    The query "Was there any party that won in all the parishes of a municipality in the district of Porto? Indicate the municipality code and the party acronym." is universal in nature. Compare, from the point of view of execution time and execution plan, the double negation and counting strategies in three different situations (only in context Z):

    a. Without a view to calculate the winner in each parish.
    b. With a view.
    c. With a materialized view (possibly with indexes).

5. Compare execution plans for the query: "How many votes did PS and PSD obtain in districts 11, 15, and 17?" considering in environment Z:
    a. Using B-tree indexes on zconcelhos.distrito e zvotacoes.partido.
    b. Using bitmap indexes.